JACOB JIBU

Electrical & Electronics Engineering Undergraduate | Chennai, Tamil Nadu
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SUMMARY

Electrical and Electronics Engineering undergraduate with hands-on experience in industrial electrical systems, hardware validation, control systems, and reliability-focused engineering. Experienced in subsystem testing, fault handling, failure prevention, electrical troubleshooting, and system integration through an industrial internship and applied project work. Strong foundation in electrical, electronic, thermal, and control-system principles, using MATLAB, Simulink, Python, and embedded platforms for engineering analysis and validation.

EDUCATION

B.Tech in Electrical and Electronics Engineering (7th Semester) | Vellore Institute of Technology, Chennai, TN 08/2023 – Present

Cumulative GPA: 8.19 / 10

Profile Focus: Core electrical engineering coursework spanning power systems, electrical machines, power electronics, control systems, and circuit design.

Higher Secondary Education | MTHSS, Kaloor, Ernakulam, Kerala 07/2021 – 03/2023

Overall Marks: 94.66% | Coursework included a focus on Electrical Systems.

EXPERIENCE

Automation & Industrial Intern | Apollo Tyres Ltd. Jun 2025 – Jul 2025

- Gained hands-on exposure to industrial electrical systems, manufacturing processes, electrical distribution networks, induction motors, electrical drives, and industrial control panels in a high-volume manufacturing environment.
- Observed preventive maintenance, equipment troubleshooting, electrical fault identification, and safety practices across production-line equipment.
- Studied production workflows and process automation with a focus on equipment reliability, operational efficiency, and maintenance practices.
- Gained introductory exposure to PLC-controlled manufacturing systems as part of the broader automation environment.

Electrical Systems Engineer, Recovery Systems | Team IGNITION Aug 2025 – May 2026

- Designed and validated electrical and actuation systems for a controllable, paraglider-based rocket recovery system, incorporating fault-handling and fail-safe mechanisms for reliable operation near 1 km apogee.
- Developed heartbeat-based wireless link monitoring and acknowledgement logic to continuously detect communication failures and trigger safe, link-loss disarm behavior.
- Implemented neutral-position correction and stall-protection logic to prevent actuator overload and improve system reliability during recovery operation.
- Coordinated hardware development, subsystem integration, testing, and reliability-focused design validation across the recovery system team; managed project timelines and allocated responsibilities.

PROJECTS

Adaptive Matrix LED Headlamp System Jan 2026 – Apr 2026

- Designed PWM-driven LED driver circuitry and a multiplexed matrix drive for an adaptive automotive headlamp, reducing digital I/O usage while controlling individual LED zones.
- Developed zone-based adaptive dimming control logic for automotive power and lighting systems, using vehicle-detection input to adjust beam pattern in real time.

Technologies: PWM • LED Driver Circuits • Automotive Power Systems • Arduino • Computer Vision (input signal)

Smart Modular Climate Regulation Platform (SIH Internal Prototype) Aug 2025 – Oct 2025

- Developed a closed-loop climate control system for agricultural cold storage (onion storage), targeting conditions of approximately 2–4°C and 60–70% relative humidity.
- Designed the power and control scheme for a Peltier (thermoelectric) cooling module, exhaust ventilation fan, and pump, integrating DHT11 temperature and humidity sensors for feedback.
- Built sensor interfacing and actuator control firmware with remote monitoring for the closed-loop system.

Technologies: Peltier Module • Control Systems • Temperature & Humidity Sensors • ESP32 • MQTT

ESP32-Based Smart Energy Management & Predictive Load Shedding System Jul 2025 – Dec 2025

- Developed an energy management system using historical load data to identify demand patterns and support predictive load-shedding decisions, applying machine learning for load forecasting.
- Designed relay-based power switching circuitry for load shedding across multiple electrical load nodes.
- Implemented real-time monitoring and mobile-controlled load management with data acquisition across load nodes.

Technologies: Relay Control • Load Forecasting • Power Distribution • ESP32 • MQTT

Hack Computer Architecture System Aug 2026 – Present

- Building a 16-bit computer from basic logic gates based on the Hack architecture, including CPU (ALU, registers, program counter, control unit), RAM/ROM, and clock generation.
- Developing a video/display subsystem with SRAM-based video memory and VGA timing generation, including sync counters and pixel-count logic.

Technologies: Digital Logic Design • Computer Architecture • VGA Timing

SKILLS

Reliability & Engineering: Hardware Validation • Failure Prevention • Fault Analysis • Preventive Maintenance • System Integration • Reliability Analysis

Core Electrical: Circuit Design, Power Electronics, Electrical Machines & Drives, Power Systems & Distribution, Control Systems, Industrial Automation, Signal Processing

Software & Design Tools: MATLAB, Simulink, LTspice, Keil µVision, Arduino IDE

Programming & Embedded Platforms: Embedded C, C, Python, Java, ESP32, ESP8266, Arduino

Additional Domains: IoT Systems, Wireless Communication, Sensor Interfacing, Machine Learning for Control Applications

Soft Skills: Teamwork, Problem Solving, Leadership, Communication, Adaptability

Languages: English (Full Professional Proficiency), Malayalam (Native), Tamil (Conversational), Hindi (Listening Proficiency)

EXTRACURRICULAR ACTIVITIES

Team IGNITION, VIT Chennai

- Participated in the design and development of experimental rocketry systems.
- Worked on recovery mechanisms, wireless communication systems, and electrical subsystem integration.

Smart India Hackathon (Internal Selection Round)

- Participated in the development of a closed-loop climate regulation system for agricultural applications.